

Appendix D Laboratory board Connection Summary

PORTA	PA0-7	8 x Toggle Switchs 8 x Push Buttons 3 x 4 Keypad (PA0-3 Row1-3 --- PA4-7 Col1-3) External Connector	
PORTB	PB0	Real time clock SPI	SS
	PB1	Real time clock SPI	SCK
	PB2	Real time clock SPI	MOSI
	PB3	Real time clock SPI	MISO
	PB4	Speaker	OC2A
	PB5	Motor Speed	OC1A
	PB6	Lamp	OC1B
	PB7	Heater	OC1C/OC0A
PORTC	PC0-7	8 x LEDs 2 x Hexadecimal 7 Segment Display External Connector	
PORTD	PD0/PD1	SCL/SDA	EEPROM (address 0x57)
	PD0/PD1	SCL/SDA	Multicolour LED (address 0x66)
	PD0/PD1	SCL/SDA	LCD (address 0x28)
	PD2	UART RX (USB)	
	PD3	UART Tx (USB)	
	PD4	Frequency input	
	PD5	External connector interface selection	
	PD6	External connector interface selection	
	PD7	Fan speed	
PORTE	PE0	Switches/Buttons/Keypad select for PORTA	
	PE1	Switches/Buttons/Keypad select for PORTA	
	PE2	Buzzer	
	PE3		
	PE4	Real time clock alarm (INT4)	
	PE5	Motion sensor (PIR) (INT5)	
	PE6	Rotary encoder Channel B (INT6)	
	PE7	Rotary encoder Channel A (INT7)	
PORTF	PF0	Light dependant resistor	ADC0
	PF1	Humidity sensor or Variable Resistor 2	ADC1
	PF2	Microphone or Variable Resistor 1	ADC2
	PF3	Temperature	ADC3
	PF4	JTAG	
	PF5	JTAG	
	PF6	JTAG	
	PF7	JTAG	